NEET UG BIOLOGY STANDARD

Biological Classification

Comprehensive Study Guide, Visual Concept Atlas, Worked Examples, and Original NEET UG Practice Assessment

1. Learning Objectives & Chapter Overview

Biological classification is the scientific procedure of arranging organisms into hierarchical groups based on their similarities and dissimilarities. For the **NEET UG** examination, this topic forms the bedrock of diversity in the living world. Questions frequently test comparative Kingdom traits, exceptions to general rules, microbial nutrition, fungal life cycles and spore types, and the molecular nature of acellular infectious agents.

Key Learning Objectives

- Trace the historical shift from instinctive classification to Aristotle's morphological system, Linnaeus's Two Kingdom system, and R.H. Whittaker's Five Kingdom system.
- Compare Monera, Protista, Fungi, Plantae, and Animalia across five core criteria: cell structure, body organisation, mode of nutrition, reproduction, and phylogenetic relationships.
- Differentiate Archaebacteria and Eubacteria, including cyanobacteria, chemosynthetic autotrophs, heterotrophic bacteria, and wall-less Mycoplasma.
- Examine the five major groups of Kingdom Protista: Chrysophytes, Dinoflagellates, Euglenoids, Slime moulds, and Protozoans.
- Master the vegetative, asexual, and sexual reproductive cycles of Kingdom Fungi and distinguish between Phycomycetes, Ascomycetes, Basidiomycetes, and Deuteromycetes.
- Understand the defining characteristics of Kingdoms Plantae and Animalia, and analyze acellular entities (Viruses, Viroids, Prions) and symbiotic Lichens.

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2. Evolution of Classification Systems

2.1 Early Attempts and Aristotle's Classification

In the earliest days of human civilisation, classification was done instinctively rather than scientifically, driven by the practical human need to use organisms for food, shelter, and clothing.

Aristotle was the earliest pioneer to attempt a more scientific basis for classification. He relied on simple morphological characters:

- **Plants:** Classified into three groups based on habit—**Trees**, **Shrubs**, and **Herbs**.
- **Animals:** Divided into two major groups—those which had **red blood** and those that **did not**.

2.2 Linnaeus's Two Kingdom System and Its Limitations

During the time of **Carolus Linnaeus**, a **Two Kingdom system of classification** was developed, dividing all living organisms into **Kingdom Plantae** (all plants) and **Kingdom Animalia** (all animals). In earlier classification schemes, bacteria, blue-green algae, fungi, mosses, ferns, gymnosperms, and angiosperms were all placed together under 'Plants' based on a single unifying character: **all of them had a cell wall in their cells**.

While easy to understand and apply at a superficial level, the Two Kingdom system proved inadequate because a large number of organisms did not fall neatly into either category, and it brought together groups that differed widely in fundamental biology:

1. **No distinction between Prokaryotes and Eukaryotes:** It grouped prokaryotic bacteria and blue-green algae (cyanobacteria) together with eukaryotic groups (mosses, ferns, gymnosperms, angiosperms).

2. **No distinction between Unicellular and Multicellular organisms:** For example, unicellular *Chlamydomonas* and multicellular filamentous *Spirogyra* were placed together under algae.
3. **No distinction between Photosynthetic and Non-photosynthetic organisms:** Heterotrophic fungi were grouped with autotrophic green plants, despite showing a fundamental difference in cell wall composition—**fungi possess chitinous cell walls**, whereas **green plants possess cellulosic cell walls**.

Consequently, biologists recognised the need to incorporate characteristics beyond gross morphology, such as **cell structure, nature of the cell wall, mode of nutrition, habitat, methods of reproduction, and evolutionary (phylogenetic) relationships**.

2.3 R.H. Whittaker's Five Kingdom Classification (1969)

In 1969, **R.H. Whittaker** proposed the **Five Kingdom Classification**, defining the kingdoms: **Monera, Protista, Fungi, Plantae, and Animalia**.

Five Main Criteria Used by R.H. Whittaker (1969)

1. **Cell structure** (Prokaryotic vs. Eukaryotic; presence/absence and composition of cell wall)
2. **Body organisation** (Cellular, multicellular/loose tissue, tissue/organ, or organ system)
3. **Mode of nutrition** (Autotrophic—photosynthetic or chemosynthetic; Heterotrophic—saprophytic, parasitic, or holozoic)
4. **Reproduction**
5. **Phylogenetic relationships** (Evolutionary relationships)

Key Reorganisations in Whittaker's System:

- All **prokaryotic** organisms were separated and grouped under **Kingdom Monera**.
- All **unicellular eukaryotic** organisms were placed in **Kingdom Protista**. Notably, Kingdom Protista brought together *Chlamydomonas* and *Chlorella* (earlier placed in Algae within Plants, and both having cell walls) with *Paramoecium* and *Amoeba* (earlier placed in the Animal kingdom, and both lacking cell walls).
- **Heterotrophic eukaryotic** organisms with chitinous cell walls were placed in a dedicated kingdom—**Kingdom Fungi**.

Note on the Three-Domain System: A three-domain system has also been proposed later, which divides Kingdom Monera into two domains while leaving all remaining eukaryotic kingdoms in the third domain, thereby resulting in a **six-kingdom classification**.

Comparative Characteristics of the Five Kingdoms (Whittaker, 1969)

Characters	Monera	Protista	Fungi	Plantae	Animalia
Cell type	Prokaryotic	Eukaryotic	Eukaryotic	Eukaryotic	Eukaryotic
Cell wall	Noncellulosic (Polysaccharide + amino acid)	Present in some	Present with chitin	Present (cellulose)	Absent
Nuclear membrane	Absent	Present	Present	Present	Present
Body organisation	Cellular	Cellular	Multicellular / loose tissue	Tissue / organ	Tissue / organ / organ system
Mode of nutrition	Autotrophic (chemosynthetic and photosynthetic) and Heterotrophic (saprophytic / parasitic)	Autotrophic (Photosynthetic) and Heterotrophic	Heterotrophic (Saprophytic / Parasitic)	Autotrophic (Photosynthetic)	Heterotrophic (Holozoic / Saprophytic etc.)

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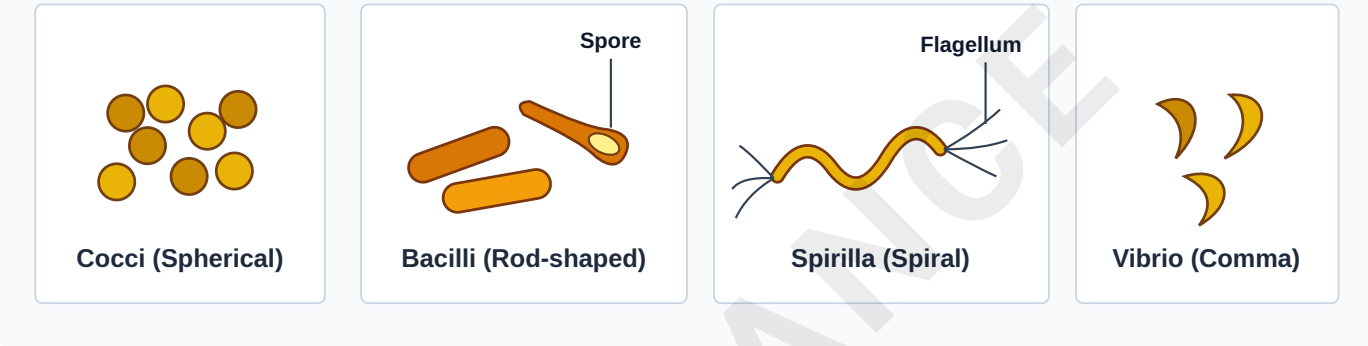
3. Kingdom Monera

Bacteria are the sole members of Kingdom Monera. They are the most abundant microorganisms on Earth and are cosmopolitan in distribution—hundreds of bacteria are present in a single handful of soil. They also thrive in extreme habitats where very few other life forms can survive, such as hot springs, deserts, snow, and deep oceans. Many live in or on other organisms as parasites.

3.1 Morphological Categories of Bacteria

Based on their shape, bacteria are grouped into four categories:

1. **Coccus** (plural: *cocci*): Spherical-shaped.
2. **Bacillus** (plural: *bacilli*): Rod-shaped (often seen with endospores under certain conditions).
3. **Vibrium** (plural: *vibrio*): Comma-shaped.
4. **Spirillum** (plural: *spirilla*): Spiral-shaped (often bearing tufts of flagella at one or both ends).



Tutorial Diagram 1: The four morphological categories of bacteria based on cell shape—spherical Cocci, rod-shaped Bacilli (showing endospore), spiral Spirilla (showing polar flagella), and comma-shaped Vibrio.

3.2 Bacterial Structure vs. Metabolic Diversity

Although bacterial structure is **very simple**, they are **very complex in behaviour**. Compared to all other groups of organisms, **bacteria as a group show the most extensive metabolic diversity**:

- **Autotrophic bacteria:** Synthesise their own organic food from inorganic substrates. They may be **photosynthetic autotrophic** or **chemosynthetic autotrophic**.
- **Heterotrophic bacteria:** The **vast majority** of bacteria are heterotrophs; they do not synthesise their own food but depend on other living organisms (parasites) or on dead organic matter (saprophytes) for nutrition.

3.3 Archaebacteria

Archaebacteria are a special group of ancient bacteria capable of surviving in some of the **most harsh habitats**:

- **Halophiles:** Live in extreme salty areas.
- **Thermoacidophiles:** Live in hot springs (high temperature and acidity).
- **Methanogens:** Live in marshy (anaerobic) areas.

> **High-Yield NEET Concept: Why do Archaebacteria survive extreme conditions?**
>
> Archaebacteria differ from other bacteria in having a **different cell wall structure**, and this specific feature is responsible for their survival in extreme conditions.
>
> **Methanogens in Ruminants:** Methanogens are present in the **gut of several ruminant animals** such as cows and buffaloes. They are responsible for the production of **methane (biogas)** from the dung of these animals.

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3.4 Eubacteria ('True Bacteria')

Eubacteria are characterised by the presence of a **rigid cell wall**, and if motile, a **flagellum**.

A. Cyanobacteria (Blue-Green Algae)

- They possess **chlorophyll *a*** (similar to green plants) and are **photosynthetic autotrophs**.
- **Habitat & Habit:** Unicellular, colonial, or filamentous; found in freshwater, marine, or terrestrial environments.
- Their colonies are generally surrounded by a **gelatinous (mucilaginous) sheath**.
- They often form dense growths called **blooms in polluted water bodies**.
- **Nitrogen Fixation:** Certain filamentous cyanobacteria fix atmospheric nitrogen in specialised, thick-walled cells called **heterocysts**. Classic examples: *Nostoc* and *Anabaena*.

B. Chemosynthetic Autotrophic Bacteria

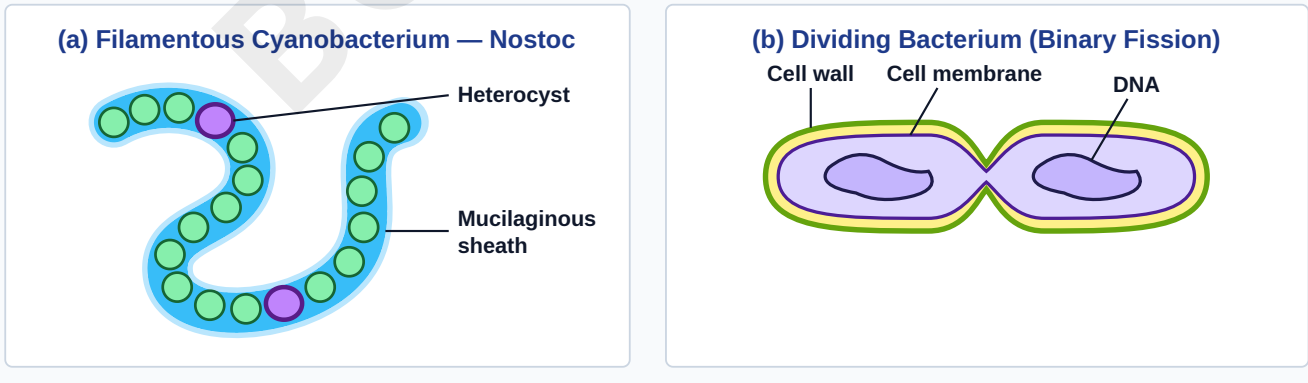
- These bacteria **oxidise various inorganic substances** such as **nitrates, nitrites, and ammonia**, and use the released chemical energy for their **ATP production**.
- They play a vital ecological role in **recycling nutrients like nitrogen, phosphorus, iron, and sulphur**.

C. Heterotrophic Bacteria

- Heterotrophic bacteria are the **most abundant in nature**, and the majority of them are important **decomposers**.
- **Beneficial Roles:** Making curd from milk (e.g., *Lactobacillus*), production of antibiotics, and fixing atmospheric nitrogen symbiotically in legume roots (e.g., *Rhizobium*).
- **Pathogenic Roles:** Cause damage to human beings, crops, farm animals, and pets. Well-known bacterial diseases include **cholera**, **typhoid**, **tetanus**, and **citrus canker**.

3.5 Reproduction in Bacteria

1. **Fission (Primary method):** Bacteria reproduce mainly by binary fission (cell division following DNA replication).
2. **Spore formation:** Under unfavourable environmental conditions, bacteria produce resistant spores.
3. **Primitive Sexual Reproduction:** They also exhibit a sort of sexual reproduction by adopting a **primitive type of DNA transfer** from one bacterium to another.



Tutorial Diagram 2: (a) A filamentous blue-green alga (*Nostoc*) showing nitrogen-fixing heterocysts and outer mucilaginous sheath. (b) A dividing bacterium undergoing binary fission showing cell wall, cell membrane, and replicated genetic material (DNA).

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3.6 Mycoplasma

> **Four Must-Know Facts About Mycoplasma for NEET UG**
>
> 1. They **completely lack a cell wall** (making them naturally resistant to cell-wall-targeting antibiotics like penicillin).
> 2. They are the **smallest living cells known**.
> 3. They **can survive without oxygen** (facultative/obligate anaerobes).
> 4. Many mycoplasma are **pathogenic in both animals and plants**.

4. Kingdom Protista

All **single-celled eukaryotes** are placed under Kingdom Protista. However, the **boundaries of this kingdom are not well defined**—what may be considered a "photosynthetic protistan" by one biologist may be considered a "plant" by another.

- **Habitat:** Members of Protista are **primarily aquatic**.
- **Evolutionary Link:** Kingdom Protista forms a **connecting link** with the kingdoms dealing with plants, animals, and fungi.
- **Cellular Structure:** Being eukaryotes, their cell body contains a **well-defined nucleus** and other **membrane-bound organelles**. Some possess flagella or cilia for locomotion.
- **Reproduction:** They reproduce both **asexually** and **sexually** by a process involving cell fusion and **zygote formation**.
- **Major Groups Included:** Chrysophytes, Dinoflagellates, Euglenoids, Slime moulds, and Protozoans.

4.1 Chrysophytes

- **Members:** Includes **diatoms** and **golden algae (desmids)**.
- **Habitat & Habit:** Found in both freshwater and marine environments. They are microscopic and **float passively in water currents (plankton)**. Most of them are photosynthetic.
- **Cell Wall Structure in Diatoms:** The cell walls form **two thin overlapping shells** that fit together like a **soap box** (an upper epitheca over a lower hypotheca). The walls are **embedded with silica**, making them **indestructible**.
- **Diatomaceous Earth:** Because their siliceous walls do not decay, diatoms have left behind vast amounts of cell wall deposits in their habitat over billions of years. This accumulation is called **diatomaceous earth**. Being gritty, this soil is used commercially in **polishing** and in the **filtration of oils and syrups**.
- **Ecological Role: Diatoms are the chief 'producers' in the oceans.**

4.2 Dinoflagellates

- **Habitat & Nutrition:** Mostly **marine and photosynthetic**.
- **Pigmentation:** They appear yellow, green, brown, blue, or red depending on the main pigments present in their cells.
- **Cell Wall:** The outer surface of the cell wall is covered with **stiff cellulose plates**.
- **Flagellation:** Most dinoflagellates have **two flagella**—one lies **longitudinally** and the other lies **transversely** in a furrow between the wall plates.
- **Red Tides:** Red dinoflagellates (Example: *Gonyaulax*) frequently undergo rapid multiplication, making the sea appear red—a phenomenon known as **red tides**. **Toxins** released by such huge numbers may kill other marine animals such as fishes.

4.3 Euglenoids

- **Habitat:** Majority are **freshwater organisms found in stagnant water**.

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- **Body Covering (No Cell Wall):** Instead of a cell wall, they possess a **protein-rich layer called a pellicle**, which makes their body **flexible**.
- **Flagellation:** They possess **two flagella**—a short one and a long one.
- **Mixotrophic Nutrition:** They are **photosynthetic in the presence of sunlight**, but when deprived of sunlight, they behave like **heterotrophs** by predating on other smaller organisms.
- **Pigments:** Remarkably, the photosynthetic pigments of euglenoids are **identical to those present in higher plants** (chlorophyll *a* and chlorophyll *b*).
- **Example:** *Euglena*.

4.4 Slime Moulds

- **Nutrition:** Slime moulds are **saprophytic protists**. Their vegetative body moves along decaying twigs and leaves, engulfing organic material.
- **Favourable Conditions (Plasmodium):** Under suitable conditions, they form a multinucleate aggregation called a **plasmodium**, which may grow and spread over several feet.
- **Unfavourable Conditions (Fruiting Bodies & Spores):** During unfavourable conditions, the plasmodium differentiates and forms **fruiting bodies bearing spores at their tips**.
- **Spore Characteristics:** The spores possess **true walls**, are **extremely resistant**, can survive for many years even under adverse conditions, and are **dispersed by air currents**.

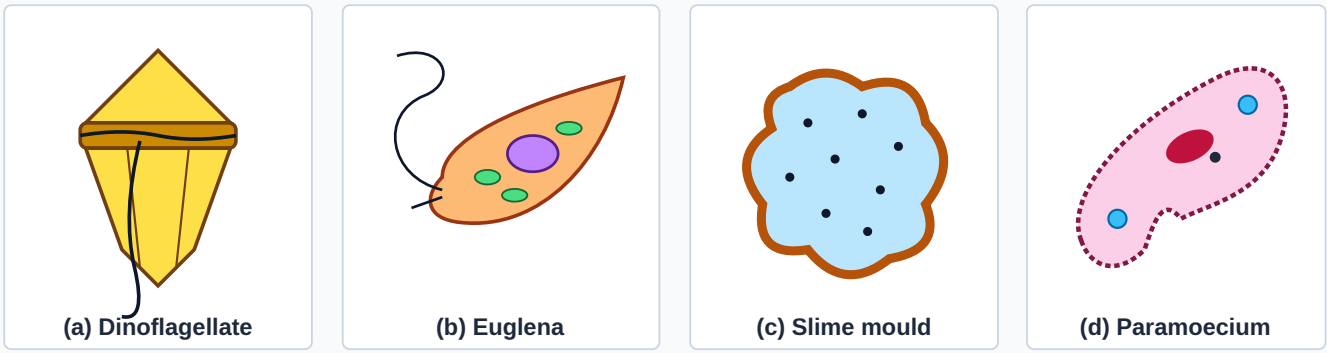
4.5 Protozoans

All protozoans are **heterotrophs** and live as **predators or parasites**. They are believed to be **primitive relatives of animals**. Based on locomotory structures and life cycle features, there are four major groups of protozoans:

Comparison of the Four Major Groups of Protozoans

Group	Habitat & Lifestyle	Locomotory / Special Features	Examples & Diseases
1. Amoeboid protozoans	Freshwater, sea water, or moist soil; free-living predators or parasites.	Move and capture prey by putting out **pseudopodia** (false feet). **Marine forms have silica shells** on their surface.	*Amoeba* (free-living); *Entamoeba* (parasitic).
2. Flagellated protozoans	Either free-living or parasitic.	Possess **flagella** for locomotion.	*Trypanosoma* (parasitic; causes **sleeping sickness**).
3. Ciliated protozoans	Aquatic, actively moving organisms.	Possess **thousands of cilia**. Have a cavity (**gullet**) opening to the outside of the cell surface; coordinated ciliary movement steers food-laden water into the gullet.	*Paramoecium*.
4. Sporozoans	Diverse parasitic organisms.	Possess an **infectious spore-like stage** in their life cycle.	*Plasmodium* (malarial parasite; causes **malaria**).

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Tutorial Diagram 3: Representative members of Kingdom Protista—(a) Dinoflagellate showing cellulose plates and two perpendicular flagella, (b) *Euglena* with flexible pellicle and two flagella, (c) Multinucleate plasmodium of a Slime mould, and (d) Ciliated protozoan *Paramoecium* showing oral groove (gullet), macronucleus, and contractile vacuoles.

5. Kingdom Fungi

Fungi constitute a unique kingdom of **heterotrophic eukaryotic organisms** showing great diversity in morphology and habitat. Familiar examples include mould on moist bread and rotten fruits, edible mushrooms, poisonous toadstools, white spots on mustard leaves (caused by the parasitic fungus *Albugo*), unicellular **yeast** (*Saccharomyces*, used to make bread and beer), wheat rust-causing *Puccinia*, and antibiotic-producing *Penicillium*.

5.1 Habitat, Structure, and Nutrition

- **Habitat:** Fungi are **cosmopolitan** and occur in air, water, soil, and on animals and plants. They prefer to grow in **warm and humid places** (which is why we store food in a refrigerator—to prevent bacterial or fungal spoilage).
- **Thallus Organisation:** With the **exception of yeasts, which are unicellular**, fungi are **filamentous**. Their bodies consist of long, slender thread-like structures called **hyphae**. The network of hyphae is known as a **mycelium**.
  - **Coenocytic (Aseptate) Hyphae:** Continuous tubes filled with multinucleated cytoplasm lacking cross walls.
  - **Septate Hyphae:** Hyphae that possess **septae or cross walls**.
- **Cell Wall Composition:** Composed of **chitin and polysaccharides**.
- **Modes of Heterotrophic Nutrition:**
  1. **Saprophytes:** Most fungi absorb soluble organic matter from dead substrates.
  2. **Parasites:** Depend on living plants and animals.
  3. **Symbionts:** Live in mutually beneficial associations—with algae as **lichens**, and with the roots of higher plants as **mycorrhiza**.

5.2 Reproduction in Fungi

- **Vegetative Reproduction:** Takes place by **fragmentation**, **fission**, and **budding**.
- **Asexual Reproduction:** Occurs through spores called **conidia**, **sporangiospores**, or **zoospores**.
- **Sexual Reproduction:** Occurs through sexual spores called **oospores**, **ascospores**, and **basidiospores**. Various spores are produced in distinct structures called **fruiting bodies**.

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The Three Sequential Steps of the Fungal Sexual Cycle

1. **Plasmogamy:** Fusion of protoplasms between two motile or non-motile gametes.
2. **Karyogamy:** Fusion of two nuclei to form a diploid ($2n$) zygotic nucleus.
3. **Meiosis in Zygote:** Reduction division in the zygote resulting in **haploid (n) spores**.

Dikaryon and Dikaryophase: When two haploid hyphae of compatible mating types fuse, in some fungi (like Phycomycetes) the fusion of two haploid cells immediately results in diploid cells ($2n$). However, in **Ascomycetes and Basidiomycetes**, an intervening **dikaryotic stage ($n + n$, i.e., two nuclei per cell)** occurs between plasmogamy and karyogamy. This condition is called a **dikaryon** and the phase is called the **dikaryophase** of the fungus. Later, the parental nuclei fuse (karyogamy) to make the cell diploid ($2n$), followed by meiosis inside fruiting bodies to produce haploid spores.

5.3 Basis of Classification of Kingdom Fungi

Kingdom Fungi is divided into four classes based on **three main criteria**:

1. **Morphology of the mycelium** (aseptate/coenocytic vs. branched and septate)
2. **Mode of spore formation** (endogenous vs. exogenous; asexual vs. sexual)
3. **Fruiting bodies** (presence, absence, and type—e.g., ascocarp, basidiocarp)

5.4 Detailed Study of the Four Fungal Classes

1. Phycomycetes (Lower / Algal Fungi)

- **Habitat:** Found in **aquatic habitats**, on decaying wood in moist and damp places, or as **obligate parasites on plants**.
- **Mycelium: Aseptate and coenocytic**.
- **Asexual Reproduction:** By **zoospores (motile)** or **aplanospores (non-motile)**. These spores are produced **endogenously inside a sporangium**.
- **Sexual Reproduction:** A diploid **zygospore** is formed by the fusion of two gametes. Gametes may be morphologically similar (**isogamous**) or dissimilar (**anisogamous or oogamous**).
- **Examples:** *Mucor*, *Rhizopus* (the bread mould), and *Albugo* (the parasitic fungus causing white rust/spots on mustard).

2. Ascomycetes (Sac-Fungi)

- **Habit:** Mostly multicellular (e.g., *Penicillium*) or rarely unicellular (e.g., yeast, *Saccharomyces*).
- **Nutrition:** Saprophytic, decomposers, parasitic, or **coprophilous (growing on dung)**.
- **Mycelium: Branched and septate**.
- **Asexual Reproduction:** By **conidia** produced **exogenously** on special mycelial stalks called **conidiophores**. Conidia germinate to produce mycelium.
- **Sexual Reproduction:** By sexual spores called **ascospores**, which are produced **endogenously** in sac-like **asci** (singular: *ascus*). These asci are arranged in fruiting bodies called **ascocarps**.
- **Examples & Importance:** *Aspergillus*, *Claviceps*, and *Neurospora* (**used extensively in biochemical and genetic work**). Many members like **morels and truffles are edible** and considered delicacies.

3. Basidiomycetes (Club Fungi)

- **Common Forms & Habitat:** Includes **mushrooms, bracket fungi, or puffballs**. They grow in soil, on logs and tree stumps, and in living plant bodies as parasites (e.g., **rusts and smuts**).
- **Mycelium: Branched and septate**.
- **Asexual Reproduction:** Asexual spores are **generally not found**, but **vegetative reproduction by fragmentation** is common.

BOSONANCE

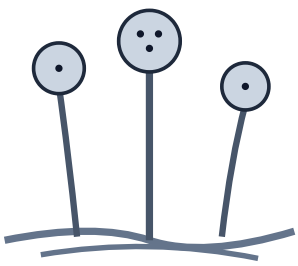
- **Sexual Reproduction: Sex organs are absent**, but **plasmogamy** is brought about by the fusion of two vegetative or somatic cells of different strains or genotypes. The resulting structure is **dikaryotic ($n + n$)**, which ultimately gives rise to the **basidium**. **Karyogamy and meiosis** take place in the basidium, producing **four basidiospores**. Basidiospores are **exogenously produced** on the basidium (plural: *basidia*), and basidia are arranged in fruiting bodies called **basidiocarps**.
- **Examples:** *Agaricus* (mushroom), *Ustilago* (smut), and *Puccinia* (rust fungus).

4. Deuteromycetes (Imperfect Fungi)

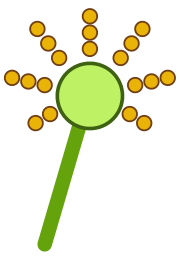
- **Why 'Imperfect Fungi'?** Only the **asexual or vegetative phases** of these fungi are known; their sexual (perfect) stages are absent or undiscovered.
- **Taxonomic Reassignment:** Once the perfect (sexual) stages of members of Deuteromycetes were discovered, they were moved into the classes they rightly belong to—most often **Ascomycetes and Basidiomycetes**.
- **Reproduction:** Reproduce **only by asexual spores known as conidia**.
- **Mycelium: Septate and branched**.
- **Nutrition & Ecological Role:** Some are saprophytes or parasites, while a **large number of them are decomposers of litter and help in mineral cycling**.
- **Examples:** *Alternaria*, *Colletotrichum*, and *Trichoderma*.

Master Comparison Table of the Four Classes of Kingdom Fungi

Feature	Phycomycetes	Ascomycetes	Basidiomycetes	Deuteromycetes
Common Name	Algal / Lower fungi	Sac-fungi	Mushrooms, bracket fungi, puffballs	Imperfect fungi
Mycelium	**Aseptate & coenocytic**	Branched & septate	Branched & septate	Branched & septate
Asexual Spores	**Zoospores** (motile) or **Aplanospores** (non-motile); **endogenous** in sporangium	**Conidia**; **exogenous** on conidiophores	**Generally absent** (vegetative by fragmentation)	**Conidia** (exogenous)
Sexual Spores	**Zygospore** (via isogamy, anisogamy, or oogamy)	**Ascospores**; **endogenous** in sac-like asci (in ascocarps)	**Basidiospores** (4 per basidium); **exogenous** on basidium (in basidiocarps)	**Absent / Not known**
Key Examples	*Mucor*, *Rhizopus*, *Albugo*	*Penicillium*, *Saccharomyces*, *Aspergillus*, *Claviceps*, *Neurospora*, Morels, Truffles	*Agaricus*, *Ustilago*, *Puccinia*	*Alternaria*, *Colletotrichum*, *Trichoderma*



(a) Mucor (Phycomycetes)



(b) Aspergillus (Ascomycetes)



(c) Agaricus (Basidiomycetes)

Tutorial Diagram 4: Representative members of Kingdom Fungi—(a) *Mucor* showing sporangiophores bearing globose sporangia, (b) *Aspergillus* showing a conidiophore with radiating chains of exogenous conidia, and (c) Basidiocarp (fruiting body) of *Agaricus* (mushroom) showing pileus (cap), gills, annulus, and stipe.

BOSONANCE

6. Overview of Kingdoms Plantae and Animalia

6.1 Kingdom Plantae

- **General Definition:** Includes all **eukaryotic chlorophyll-containing organisms** commonly called plants.
- **Cellular Structure:** Eukaryotic cells with prominent **chloroplasts** and a cell wall mainly made of **cellulose**.
- **Exceptions in Nutrition (Partially Heterotrophic Plants):**
  - **Insectivorous plants: Bladderwort** (*Utricularia*) and **Venus fly trap** (*Dionaea*).
  - **Parasitic plants:** *Cuscuta* (dodder).
- **Major Groups Included:** Algae, Bryophytes, Pteridophytes, Gymnosperms, and Angiosperms.
- **Alternation of Generations:** The life cycle of plants exhibits two distinct phases—the **diploid ($2n$) sporophytic phase** and the **haploid (n) gametophytic phase**—that alternate with each other. The relative lengths of the haploid and diploid phases, and whether these phases are free-living or dependent on others, vary among different plant groups.

6.2 Kingdom Animalia

- **General Definition:** Characterised by **heterotrophic, eukaryotic, multicellular organisms** whose cells **completely lack cell walls**.
- **Nutrition & Digestion:** They directly or indirectly depend on plants for food. They digest their food in an **internal cavity** and store food reserves as **glycogen or fat**. Their mode of nutrition is **holozoic** (by ingestion of food).
- **Growth & Coordination:** They follow a definite growth pattern and grow into adults that have a definite shape and size. Higher forms exhibit elaborate **sensory and neuromotor mechanisms**, and most are capable of locomotion.
- **Reproduction:** Sexual reproduction occurs by copulation of male and female followed by **embryological development**.

7. Viruses, Viroids, Prions, and Lichens

In R.H. Whittaker's Five Kingdom classification, there is **no mention of lichens and acellular organisms like viruses, viroids, and prions**. Viruses were excluded because they are **not considered truly 'living'** if we define living organisms as those possessing a cellular structure.

7.1 Viruses

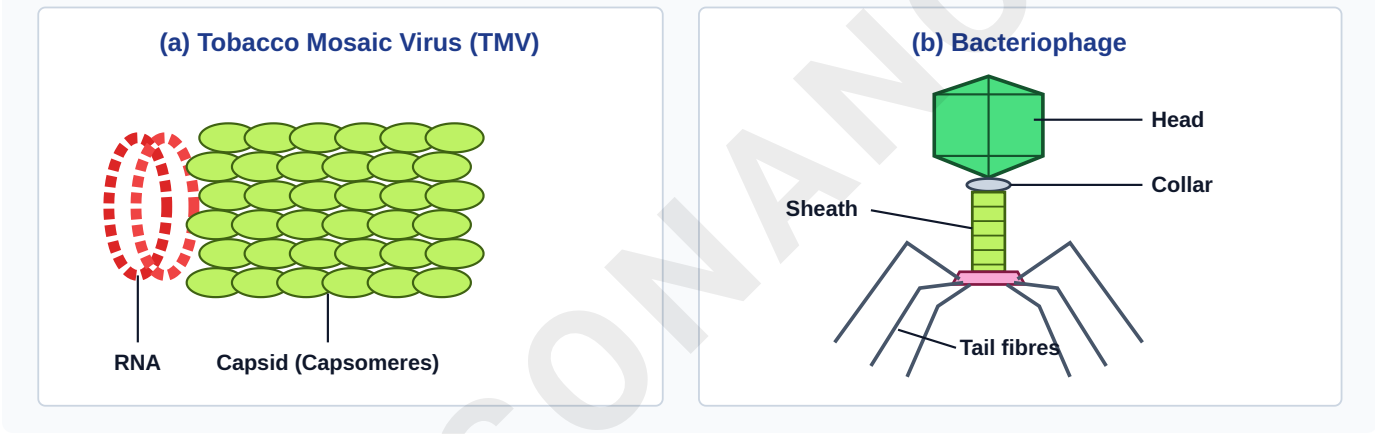
- **Non-cellular & Obligate Parasites:** Viruses are non-cellular organisms characterised by having an **inert crystalline structure outside the living cell**. Once they infect a cell, they take over the biosynthetic machinery of the host cell to replicate themselves, killing the host. Thus, **viruses are obligate parasites**.

> **Historical Milestones in Virology (High-Yield for NEET Matching Questions)**
>
> - **Meaning of 'Virus':** Means **venom or poisonous fluid**.
> - **Dmitri Ivanowsky (1892):** Recognised certain microbes as the causal organism of the **mosaic disease of tobacco**. He found that they were **smaller than bacteria because they passed through bacteria-proof filters**.
> - **M.W. Beijerinck (1898):** Demonstrated that the extract of infected tobacco plants could cause infection in healthy plants; he named the new pathogen "virus" and called the fluid *Contagium vivum fluidum* (**infectious living fluid**).
> - **W.M. Stanley (1935):** Showed that **viruses could be crystallised** and that these **crystals consist largely of proteins**.

BOSONANCE

Molecular Structure and Genetic Material of Viruses

- **Nucleoprotein Nature:** A virus is a **nucleoprotein** (composed of protein + nucleic acid), and its **genetic material is infectious**.
- **Rule of Nucleic Acids:** In addition to proteins, viruses contain genetic material that can be **either RNA or DNA**. **No virus contains both RNA and DNA.**
  - **Viruses that infect plants:** Generally have **single-stranded RNA (ssRNA)**.
  - **Viruses that infect animals:** Have either **single- or double-stranded RNA (ssRNA or dsRNA)** or **double-stranded DNA (dsDNA)**.
  - **Bacterial viruses or Bacteriophages** (viruses that infect bacteria): Usually **double-stranded DNA (dsDNA) viruses**.
- **Capsid & Capsomeres:** The protein coat called the **capsid**, made of small subunits called **capsomeres**, protects the nucleic acid. These capsomeres are arranged in **helical** (e.g., Tobacco Mosaic Virus) or **polyhedral** geometric forms (e.g., Bacteriophage head).
- **Viral Diseases in Humans:** Mumps, smallpox, herpes, influenza, common cold ('flu'), and **AIDS**.
- **Viral Symptoms in Plants:** Mosaic formation, leaf rolling and curling, yellowing and vein clearing, dwarfing, and stunted growth.



Tutorial Diagram 5: Structure of Viruses—(a) Tobacco Mosaic Virus (TMV) showing helical capsid composed of capsomeres surrounding coiled single-stranded RNA, and (b) Bacteriophage showing polyhedral head, collar, contractile sheath, base plate, and tail fibres.

7.2 Viroids

- **Discovery:** Discovered in **1971 by T.O. Diener**.
- **Size & Composition:** Smaller than viruses; consists of a **free RNA of low molecular weight** and **completely lacks the protein coat** (capsid) found in viruses—hence the name *viroid*.
- **Disease Caused:** Causes **potato spindle tuber disease**.

7.3 Prions

- **Nature & Size:** Infectious agents transmitted in certain infectious neurological diseases, consisting of **abnormally folded protein** (lacking any nucleic acid) and **similar in size to viruses**.
- **Notable Diseases Caused by Prions:**
  1. **Bovine Spongiform Encephalopathy (BSE)**, commonly called **mad cow disease** in cattle.
  2. Its analogous variant in humans: **Creutzfeldt–Jacob disease (CJD)**.

BOSONANCE

Comparison of Acellular Infectious Agents: Viruses, Viroids, and Prions

Feature	Viruses	Viroids	Prions
Chemical Composition	Nucleoprotein (Protein coat + either DNA or RNA)	Free RNA only (Low molecular weight)	Abnormally folded protein only
Protein Coat (Capsid)	Present (made of capsomeres)	Absent	Composed entirely of infectious protein
Relative Size	Smaller than bacteria (pass through bacteria-proof filters)	Smaller than viruses	Similar in size to viruses
Key Diseases	Mosaic disease of tobacco, AIDS, mumps, smallpox, herpes, influenza	Potato spindle tuber disease	BSE (mad cow disease) in cattle; CJD in humans

7.4 Lichens

- **Symbiotic Association:** Lichens are mutually useful symbiotic associations between **algae and fungi**.
- **Two Components:**
  - **Phycobiont (Algal component): Autotrophic**; prepares organic food via photosynthesis for the fungus.
  - **Mycobiont (Fungal component): Heterotrophic**; provides shelter and absorbs mineral nutrients and water for its algal partner.
- **Ecological Indicator:** Lichens are **very good pollution indicators—they do not grow in polluted areas** (especially areas polluted with sulphur dioxide).

8. Worked Conceptual Examples & Textbook Problem Walkthroughs

The following worked examples address core conceptual applications and analytical questions essential for mastering the chapter.

Worked Example 1: Economic Importance of Heterotrophic Bacteria & Archaebacteria

Problem: State two economically important uses of (a) heterotrophic bacteria and (b) archaebacteria.

Solution:

- **(a) Uses of Heterotrophic Bacteria:**
  1. Conversion of milk into **curd** and production of **antibiotics**.
  2. Symbiotic **nitrogen fixation in the roots of leguminous plants**, enriching soil fertility, as well as acting as natural decomposers.
- **(b) Uses of Archaebacteria:**
  1. **Methanogens** present in the gut of ruminant animals (cows and buffaloes) produce **methane gas (biogas)** from animal dung, which is used as a clean domestic fuel.
  2. **Thermoacidophiles** provide heat-stable enzymes capable of functioning at high temperatures and acidic conditions in biotechnology.

BOSONANCE

Worked Example 2: Ecological Significance of 'Algal Bloom' and 'Red Tides'

Problem: Distinguish between the terms 'algal bloom' and 'red tides' with respect to their causative organisms and ecological effects.

Solution:

- **Algal Bloom:** Refers to the rapid, excessive proliferation of photosynthetic microorganisms—particularly **cyanobacteria (blue-green algae)**—in nutrient-polluted freshwater or marine bodies, covering the water surface and depleting dissolved oxygen.
- **Red Tides:** Refers specifically to the rapid multiplication of **red dinoflagellates (such as *Gonyaulax*, Kingdom Protista)** in marine waters, making the sea appear red. These large populations release potent **toxins** into the water that kill fishes and other marine animals.

Worked Example 3: Ploidy Analysis in the Fungal Life Cycle

Problem: In a mushroom (*Agaricus*, class Basidiomycetes), determine the ploidy level (n, $n+n$, or $2n$) of the following structures: (i) Primary somatic hypha, (ii) Mycelium formed immediately after plasmogamy, (iii) Basidium just after karyogamy, and (iv) Basidiospores.

Solution:

1. **Primary somatic hypha:** Haploid (n).
2. **Mycelium formed immediately after plasmogamy:** Dikaryotic ($n+n$, two haploid nuclei per cell), representing the extended dikaryophase characteristic of Basidiomycetes.
3. **Basidium just after karyogamy:** Diploid ($2n$), formed when the two nuclei fuse inside the club-shaped basidium.
4. **Basidiospores:** Haploid (n), as meiosis immediately follows karyogamy inside the basidium to produce four exogenous haploid basidiospores.

Worked Example 4: Are Viruses Living or Non-Living?

Problem: Justify why viruses occupy the borderline between living and non-living entities.

Solution:

- **Non-living / Inert Properties:** Viruses lack a cellular structure (proplams, cell membrane, and metabolic enzymes). Outside a specific living host cell, they exist as **inert crystalline structures** and can be crystallised like chemicals (demonstrated by W.M. Stanley, 1935).
- **Living / Biological Properties:** They possess genetic material (either DNA or RNA) enclosed in a protein capsid. Once inside a specific host cell, the infectious genetic material takes over the host's biosynthetic machinery to **replicate** and produce progeny virions.

BOSONANCE

9. NEET UG Practice Questions with Complete Worked Solutions

Note: All questions below are original practice items designed strictly to the NEET UG examination standard to test conceptual accuracy, statement evaluation, assertion-reason logic, and matrix matching.

Practice Question 1 [Single-Choice MCQ — History & Systems of Classification]

In Whittaker's Five Kingdom classification, which of the following pairs of organisms were brought together into the **same kingdom** despite having been placed in two completely different kingdoms (Plantae and Animalia) under earlier classification systems?

(A) *Chlamydomonas* and *Spirogyra*

(B) *Chlorella* and *Amoeba*

(C) *Nostoc* and *Chlamydomonas*

(D) *Agaricus* and *Chlorella*

Correct Answer: (B)

Worked Solution: Under earlier classification systems based on the presence or absence of a cell wall, unicellular cell-walled *Chlamydomonas* and *Chlorella* were placed under Algae within Kingdom Plantae, whereas unicellular wall-less *Paramoecium* and *Amoeba* were placed in Kingdom Animalia. Whittaker (1969) placed all **unicellular eukaryotes** into **Kingdom Protista**, thereby bringing *Chlorella* and *Amoeba* into the same kingdom. (Option A includes multicellular *Spirogyra*, which remains in Plantae; Option C includes prokaryotic *Nostoc* in Monera; Option D includes *Agaricus* in Fungi).

BOSONANCE

Practice Question 2 [Multi-Statement Evaluation — Five Kingdom Characteristics]

Read the following five statements (I–V) regarding the characteristics of Whittaker's Five Kingdoms:

I. The cell wall in Kingdom Monera is non-cellulosic and composed of polysaccharide and amino acid.
II. All members of Kingdom Protista possess a well-defined cell wall and nuclear membrane.
III. Body organisation in Kingdom Fungi is multicellular / loose tissue.
IV. Chemosynthetic autotrophic nutrition is found in Kingdom Monera as well as Kingdom Protista.
V. Members of Kingdom Animalia lack cell walls and exhibit holozoic nutrition.

Which of the above statements are **correct**?

(A) I, III, and V only

(B) I, II, III, and V only

(C) II, III, and IV only

(D) I, III, IV, and V only

Correct Answer: (A)

Worked Solution:

- **Statement I is correct:** Moneran cell walls (peptidoglycan/murein) are non-cellulosic, composed of polysaccharides and amino acids.
- **Statement II is incorrect:** While all protists have a nuclear membrane (eukaryotic), a cell wall is present *only in some* (e.g., diatoms and dinoflagellates have cell walls, whereas euglenoids and protozoans lack cell walls).
- **Statement III is correct:** Fungi show multicellular / loose tissue level of body organisation.
- **Statement IV is incorrect:** Chemosynthetic autotrophic nutrition is exclusive to Kingdom Monera; autotrophic protists are strictly photosynthetic.
- **Statement V is correct:** Animalia cells lack cell walls and exhibit holozoic nutrition by ingestion.

BOSONANCE

Practice Question 3 [Assertion–Reason — Archaebacteria]

Assertion (A): Archaebacteria are capable of surviving in extreme habitats such as salty areas (halophiles), hot springs (thermoacidophiles), and marshy areas (methanogens).

Reason (R): Archaebacteria completely lack a cell wall, which prevents osmotic lysis in harsh environments.

(A) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(C) (A) is true, but (R) is false.

(D) Both (A) and (R) are false.

Correct Answer: (C)

Worked Solution: Assertion (A) is true. However, Reason (R) is false because Archaebacteria **do possess a cell wall**; they differ from other bacteria in having a **different cell wall structure**, and it is this unique cell wall structure that is responsible for their survival in extreme conditions. It is *Mycoplasma* (not Archaebacteria) that completely lack a cell wall.

Practice Question 4 [Single-Choice MCQ — Eubacteria & Cyanobacteria]

Select the **incorrect** statement regarding Eubacteria and Mycoplasma:

(A) Filamentous cyanobacteria such as *Nostoc* and *Anabaena* fix atmospheric nitrogen in specialised cells called heterocysts.

(B) Chemosynthetic autotrophic bacteria oxidise inorganic substances like nitrates, nitrites, and ammonia to release energy for ATP production.

(C) Photosynthetic autotrophic bacteria are the most abundant bacteria in nature and act as primary decomposers.

(D) Mycoplasma are the smallest living cells known, completely lack a cell wall, and can survive without oxygen.

Correct Answer: (C)

Worked Solution: Statement (C) is incorrect because **heterotrophic bacteria** (not photosynthetic autotrophic bacteria) are the most abundant in nature, and the majority of them are important decomposers. Statements (A), (B), and (D) are completely accurate facts.

BOSONANCE

Practice Question 5 [Single-Choice MCQ — Chrysophytes & Dinoflagellates]

Which of the following correctly compares **Diatoms** and **Dinoflagellates**?

(A) Diatoms have stiff cellulose plates on their outer surface, whereas Dinoflagellates have soap-box-like shells embedded with silica.

(B) Diatoms float passively in water currents (plankton) and have siliceous overlapping shells, whereas most Dinoflagellates possess two flagella (one longitudinal and one transverse) and stiff outer cellulose plates.

(C) Diatoms cause red tides in marine waters, whereas Dinoflagellates form diatomaceous earth used in filtration of oils.

(D) Both Diatoms and Dinoflagellates lack cell walls and possess a protein-rich flexible pellicle instead.

Correct Answer: (B)

Worked Solution: Diatoms (Chrysophytes) are microscopic plankton that float passively in water currents; their cell walls form two thin overlapping shells (like a soap box) embedded with silica, forming indestructible 'diatomaceous earth'. Dinoflagellates have stiff cellulose plates on the outer surface of their cell wall and typically possess two flagella—one longitudinal and one transverse in a furrow between the wall plates.

BOSONANCE

Practice Question 6 [Multi-Statement Evaluation — Euglenoids & Slime Moulds]

Consider the following statements regarding Euglenoids and Slime Moulds:

I. Euglenoids are mostly marine organisms that possess a stiff cellulose cell wall.

II. The pigments of euglenoids are identical to those present in higher plants.

III. When deprived of sunlight, euglenoids behave like heterotrophs by predating on other smaller organisms.

IV. Under unfavourable conditions, saprophytic slime moulds form an aggregation called a plasmodium that spreads over several feet.

V. Spores of slime moulds possess true walls, are extremely resistant, and are dispersed by air currents.

Which of the above statements are **correct**?

(A) II, III, and V only

(B) I, II, and IV only

(C) II, III, IV, and V only

(D) I, III, and V only

Correct Answer: (A)

Worked Solution:

- **Statement I is incorrect:** Majority of euglenoids are **freshwater** organisms found in stagnant water, and instead of a cell wall, they have a protein-rich **pellicle**.
- **Statement II & III are correct:** Euglenoids are mixotrophic (photosynthetic in sunlight, predatory heterotrophs in dark) and have pigments identical to higher plants.
- **Statement IV is incorrect:** Slime moulds form the aggregation called **plasmodium under suitable (favourable) conditions**. During **unfavourable conditions**, the plasmodium differentiates to form fruiting bodies bearing spores at their tips.
- **Statement V is correct:** Slime mould spores have true walls, are extremely resistant, and are dispersed by air currents.

BOSONANCE

Practice Question 7 [Matrix Matching — Protozoan Groups]

Match Column I (Protozoan Group / Organism) with Column II (Characteristic Feature / Disease):

Column I	Column II
(a) Marine Amoeboid protozoans	(i) Infectious spore-like stage in life cycle
(b) *Trypanosoma*	(ii) Coordinated ciliary movement steers food into gullet
(c) *Paramoecium*	(iii) Silica shells on cell surface
(d) *Plasmodium*	(iv) Flagellated protozoan causing sleeping sickness

Choose the correct option:

(A) (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)

(B) (a)-(ii), (b)-(iv), (c)-(iii), (d)-(i)

(C) (a)-(iii), (b)-(i), (c)-(ii), (d)-(iv)

(D) (a)-(iv), (b)-(iii), (c)-(i), (d)-(ii)

Correct Answer: (A)

Worked Solution:

- **(a) Marine Amoeboid protozoans — (iii):** Have silica shells on their surface.
- **(b) *Trypanosoma* — (iv):** A parasitic flagellated protozoan that causes sleeping sickness.
- **(c) *Paramoecium* — (ii):** A ciliated protozoan with thousands of cilia whose coordinated movement steers water laden with food into the cavity (gullet).
- **(d) *Plasmodium* — (i):** A sporozoan (malarial parasite) characterised by an infectious spore-like stage in its life cycle.

BOSONANCE

Practice Question 8 [Single-Choice MCQ — Fungal Sexual Cycle & Spores]

In which of the following pairs of fungal classes does an intervening **dikaryotic stage ($n + n$)** occur during the sexual cycle, and how do their sexual spores differ in their mode of formation?

(A) Phycomycetes and Ascomycetes; both produce sexual spores endogenously.

(B) Ascomycetes and Basidiomycetes; ascospores are produced endogenously in asci, whereas basidiospores are produced exogenously on basidia.

(C) Ascomycetes and Basidiomycetes; ascospores are produced exogenously, whereas basidiospores are produced endogenously.

(D) Basidiomycetes and Deuteromycetes; both produce conidia exogenously.

Correct Answer: (B)

Worked Solution: An intervening dikaryotic stage ($n + n$, two nuclei per cell, called a dikaryon / dikaryophase) occurs specifically in **Ascomycetes and Basidiomycetes**. In Ascomycetes, sexual spores (**ascospores**) are produced **endogenously** inside sac-like asci (whereas their asexual conidia are exogenous). In Basidiomycetes, four sexual spores (**basidiospores**) are produced **exogenously** on the club-shaped basidium following karyogamy and meiosis.

BOSONANCE

Practice Question 9 [Single-Choice MCQ — Identification of Fungal Classes & Examples]

A student examines four fungal specimens (P, Q, R, and S) and records the following features:

- **Specimen P:** Causes white spots on mustard leaves; possesses aseptate, coenocytic mycelium.
- **Specimen Q:** Used extensively in biochemical and genetic work; possesses branched, septate mycelium and endogenous sexual spores in sac-like asci.
- **Specimen R:** Lacks sex organs and asexual spores; reproduces sexually via somatogamy forming a basidiocarp.
- **Specimen S:** Decomposer of litter helping in mineral cycling; reproduces exclusively by asexual conidia as sexual stage is unknown.

Identify P, Q, R, and S correctly:

(A) P = *Puccinia*, Q = *Aspergillus*, R = *Agaricus*, S = *Trichoderma*

(B) P = *Albugo*, Q = *Neurospora*, R = *Ustilago*, S = *Colletotrichum*

(C) P = *Rhizopus*, Q = *Claviceps*, R = *Alternaria*, S = *Penicillium*

(D) P = *Albugo*, Q = *Neurospora*, R = *Mucor*, S = *Alternaria*

Correct Answer: (B)

Worked Solution:

- **P is *Albugo*** (Phycomycetes): Parasitic fungus on mustard with aseptate, coenocytic mycelium.
- **Q is *Neurospora*** (Ascomycetes): Extensively used in biochemical and genetic work; has septate mycelium and endogenous ascospores.
- **R is *Ustilago* (or *Agaricus* / *Puccinia*)** (Basidiomycetes): Lacks sex organs and asexual spores; shows somatogamy and basidiocarp formation.
- **S is *Colletotrichum* (or *Alternaria* / *Trichoderma*)** (Deuteromycetes): Imperfect fungus reproducing only by asexual conidia.

BOSONANCE

Practice Question 10 [Assertion–Reason — Deuteromycetes]

Assertion (A): Deuteromycetes are commonly known as 'imperfect fungi', and many of their former members have been moved to Ascomycetes and Basidiomycetes over time.

Reason (R): Only the asexual or vegetative phases of Deuteromycetes were initially known; once their sexual (perfect) stages were discovered, they were reassigned to their appropriate classes.

(A) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(C) (A) is true, but (R) is false.

(D) Both (A) and (R) are false.

Correct Answer: (A)

Worked Solution: Both Assertion and Reason are true, and Reason directly explains Assertion. Deuteromycetes are termed 'imperfect fungi' because only their asexual (conidial) or vegetative stages are known. Whenever the sexual ('perfect') stage of a fungus in this group is discovered (such as ascospore or basidiospore formation), taxonomic linkages are established and the fungus is moved out of Deuteromycetes into Ascomycetes or Basidiomycetes.

BOSONANCE

Practice Question 11 [Matrix Matching — Virology Pioneers & Acellular Agents]

Match Column I (Scientist / Infectious Agent) with Column II (Discovery / Characteristic):

Column I	Column II
(a) M.W. Beijerinck (1898)	(i) Discovered free low-molecular-weight RNA causing potato spindle tuber disease
(b) W.M. Stanley (1935)	(ii) Abnormally folded protein causing BSE in cattle and CJD in humans
(c) T.O. Diener (1971)	(iii) Called the infectious tobacco extract *Contagium vivum fluidum*
(d) Prions	(iv) Showed that viruses could be crystallised and crystals consist largely of proteins

Choose the correct option:

(A) (a)-(iv), (b)-(iii), (c)-(i), (d)-(ii)

(B) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)

(C) (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)

(D) (a)-(i), (b)-(iv), (c)-(iii), (d)-(ii)

Correct Answer: (B)

Worked Solution:

- **(a) M.W. Beijerinck (1898) — (iii):** Coined the name "virus" and called the infectious tobacco fluid *Contagium vivum fluidum* (infectious living fluid).
- **(b) W.M. Stanley (1935) — (iv):** Crystallised viruses (TMV) and showed that the crystals consist largely of proteins.
- **(c) T.O. Diener (1971) — (i):** Discovered **viroids** (free RNA lacking a protein coat, of low molecular weight, causing potato spindle tuber disease).
- **(d) Prions — (ii):** Infectious neurological agents composed of abnormally folded protein (similar in size to viruses) causing Bovine Spongiform Encephalopathy (BSE / mad cow disease) in cattle and Creutzfeldt–Jacob disease (CJD) in humans.

BOSONANCE

Practice Question 12 [Multi-Statement Evaluation — Viral Genetic Material & Lichens]

Identify the **correct** statements from the following:

I. A single virus particle can contain both DNA and RNA as its genetic material.

II. In general, viruses that infect plants have single-stranded RNA, whereas bacteriophages are usually double-stranded DNA viruses.

III. In a lichen, the heterotrophic algal component is called the phycobiont and the autotrophic fungal component is called the mycobiont.

IV. Lichens are excellent pollution indicators because they flourish luxuriously in industrial areas with high sulphur dioxide pollution.

V. Bladderwort and Venus fly trap are examples of partially heterotrophic insectivorous plants, whereas *Cuscuta* is a parasitic plant.

(A) II and V only

(B) I, II, and V only

(C) II, III, and IV only

(D) II, IV, and V only

Correct Answer: (A)

Worked Solution:

- **Statement I is incorrect:** A virus contains either RNA or DNA; **no virus contains both RNA and DNA**.
- **Statement II is correct:** Plant viruses generally possess single-stranded RNA (ssRNA), while bacteriophages are usually double-stranded DNA (dsDNA) viruses.
- **Statement III is incorrect:** In lichens, the algal component (**phycobiont**) is **autotrophic**, whereas the fungal component (**mycobiont**) is **heterotrophic**.
- **Statement IV is incorrect:** Lichens are pollution indicators because they **do not grow in polluted areas**.
- **Statement V is correct:** Bladderwort and Venus fly trap are insectivorous plants, and *Cuscuta* is a parasite; all three belong to Kingdom Plantae.

BOSONANCE

10. Common NEET UG Pitfalls & Rapid Revision Checklist

Watch Out for These 7 Classic NEET Exam Traps!

1. **Endogenous vs. Exogenous Fungal Spores:**
   - **Endogenous** (formed *inside* a sac/sporangium): **Zoospores & Aplanospores** (in sporangium of Phycomycetes) and **Ascospores** (in ascus of Ascomycetes).
   - **Exogenous** (formed *externally* on stalks/basidia): **Conidia** (on conidiophores in Ascomycetes & Deuteromycetes) and **Basidiospores** (on basidium in Basidiomycetes).
2. **Unicellular Fungus Exception:** While almost all fungi are filamentous (hyphal), **yeast (*Saccharomyces*) is unicellular** and belongs to class **Ascomycetes**.
3. **Silica in Protista:** Silica is found in **two** places mentioned in this chapter—(i) embedded in the soap-box cell walls of **Diatoms (Chrysophytes)**, and (ii) as surface silica shells in **marine Amoeboid protozoans**.
4. **Plasmodium (Slime Mould) vs. *Plasmodium* (Sporozoan):**
   - **plasmodium** (lowercase, non-italicised): The multinucleate vegetative aggregation formed by saprophytic **slime moulds** under favourable conditions.
   - ***Plasmodium*** (capitalised, italicised genus): The parasitic **sporozoan protozoan** that causes malaria.
5. **Cell Wall Exceptions Across Kingdoms:**
   - In **Monera**: Most have a non-cellulosic peptidoglycan wall, Archaebacteria have a distinct wall structure, and **Mycoplasma completely lack a cell wall**.
   - In **Protista**: Diatoms (siliceous wall), Dinoflagellates (cellulose plates), and Slime mould spores (true walls) have walls, whereas **Euglenoids (protein pellicle)** and most **Protozoans** lack cell walls.
6. **Scientist Chronology in Virology: Ivanowsky (1892)** discovered filterable microbes causing tobacco mosaic disease → **Beijerinck (1898)** coined *Contagium vivum fluidum* and named "virus" → **Stanley (1935)** crystallised viruses → **Diener (1971)** discovered viroids.
7. **Phycobiont vs. Mycobiont:** Always associate **Phyco = Algae (Autotrophic)** and **Myco = Fungi (Heterotrophic)**.

Final One-Page Summary & Revision Checklist

- **Aristotle:** Plants (trees, shrubs, herbs); Animals (with red blood vs. without red blood).
- **Whittaker (1969):** Five kingdoms (Monera, Protista, Fungi, Plantae, Animalia) based on cell structure, body organisation, mode of nutrition, reproduction, and phylogenetic relationships.
- **Monera:** Solely bacteria; most extensive metabolic diversity. Archaebacteria (halophiles, thermoacidophiles, methanogens in ruminant gut). Eubacteria: Cyanobacteria (chlorophyll *a*, gelatinous sheath, heterocysts in *Nostoc* & *Anabaena*), Chemosynthetic autotrophs (oxidise nitrates, nitrites, ammonia), Heterotrophs (decomposers, curd, antibiotics, legume N_2-fixation; pathogens of cholera, typhoid, tetanus, citrus canker), and wall-less anaerobic *Mycoplasma*.
- **Protista:** Single-celled aquatic eukaryotes. Chrysophytes (diatoms + desmids; chief ocean producers; diatomaceous earth), Dinoflagellates (2 flagella, cellulose plates; *Gonyaulax* causes red tides), Euglenoids (*Euglena*; pellicle, mixotrophic, higher-plant pigments), Slime moulds (plasmodium in favourable, fruiting bodies with true-walled spores in unfavourable), Protozoans (Amoeboid, Flagellated, Ciliated, Sporozoans).
- **Fungi:** Chitinous walls; saprophytes, parasites, symbionts (lichens, mycorrhiza). Sexual cycle: Plasmogamy → Karyogamy → Meiosis. Classes: Phycomycetes (*Mucor*, *Rhizopus*, *Albugo*), Ascomycetes (*Penicillium*, *Saccharomyces*, *Aspergillus*, *Claviceps*, *Neurospora*, morels, truffles), Basidiomycetes (*Agaricus*, *Ustilago*, *Puccinia*), Deuteromycetes (*Alternaria*, *Colletotrichum*, *Trichoderma*).

BOSONANCE

- **Acellular Agents & Lichens:** Viruses (obligate intracellular nucleoprotein parasites; either DNA or RNA), Viroids (free low-MW RNA, no capsid; potato spindle tuber disease), Prions (abnormally folded protein; BSE & CJD), Lichens (symbiotic phycobiont + mycobiont; SO_2 pollution indicators).

BOSONANCE